

COMPUTER NETWORKS

Unit wise Solutions:

Computer Networks:

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Introduction to Computer Network

① Explain about Connection-oriented network services.

(2081)

→ A connection-oriented network services is a type of communication where a proper connection is first made between the sender and receiver before any data is sent. After the connection is set up, data is transferred and finally the connection is closed.

This type of service is used in TCP (Transmission Control Protocol).

How it works?

1. First a connection is established between sender & receiver.
2. Then data is sent through that connection in ordered way.
3. After the data transfer is completed, the connection is closed.

eg:

When you open a website your browser first connects to the server. Only after this connection is made, the web page data is sent to your device.

② Difference between web server and proxy server. (2081)

web server

- A web server stores website files and sends them to users when requested.
- It directly provides web pages, images and data to the client.
- It is the final source of content.
- e.g. Apache, Nginx

proxy server

- A proxy server acts as a middle layer between the user and the web server.
- It forwards requests from users to web servers and returns the response.
- It is not the original source of content.
- e.g. Squid proxy

③ Explain different types of network topologies.

→ A network topology is the way computers and devices are arranged and connected in a network.

Bus topology:

→ In bus topology, all devices are connected to a single main cable called the bus.

- easy to set up and cheap

- If the main cable fails, the whole network stops.

e.g. small office network

Star topology:

→ In star topology all devices are connected to a central device like a hub or switch.

- easy to manage and trouble shoot
- If the central device fails, the network stops.

e.g. most modern LAN networks

Ring topology:

→ In Ring topology, each device is connected to two other devices, forming a ring shape.

- Data travels in one direction or both directions.
- If one device fails, it can affect the whole network.

mesh topology:

→ In mesh topology, every device is connected to every other device.

- very reliable
- expensive and complex.

e.g. Internet backbone networks

Tree topology:

→ Tree topology is a combination of star and bus topology. It has a hierarchical structure.

- easy to expand
- If the main backbone fails, large part of network is affected.

④ Explain different types of network types:

(2080)

→ A computer network can be classified into different types based on its size, coverage area and purpose.

1 LAN (Local Area network)

→ A LAN connects computer within a small area like a room, building or school.

- High speed
- low cost
- privately owned,

e.g. computer lab in a school.

2 MAN (Metropolitan Area network)

→ A MAN covers a large area than LAN such as a city.

- connects multiple LANs
- moderate speed
- used in city-wide networks

e.g. network between branches of bank in a city.

3 WAN (Wide Area network)

→ A WAN covers a very large area like countries or continents.

- uses public networks like Internet
- slower than LAN
- very large coverage

e.g. the Internet

4 PAN (Personal Area Network)

→ A pan is a very small network used for personal devices.

- very short range (a few meters)
- wireless or wired
- low power

e.g. mobile connected to bluetooth headphones

5 write short notes on (any two)

a. protocol and standards

b. Switch

c. Check Sum

2079

a) protocol and standards:

→ A protocol is a set of rules that defines how data is transmitted and received between devices in a network. It ensures proper communication between sender and receiver.

e.g. TCP, IP, HTTP, FTP.

Standards:

→ Standards are agreed rules or guidelines that help different devices and systems communicate with each other in a uniform way. Standards make networks compatible and reliable.

e.g. IEEE standards, ISO standards,

Conclusion:

Protocols defines how communication happens, while Standards ensure compatibility between different systems.

2 Switch:

→ A Switch is a networking device used to connect multiple devices in a LAN. It receives data and sends it only to the intended device instead of broadcasting to all devices.

Features:

- Works at data link layer (layer 2)
- Reduces network traffic
- Improves speed & efficiency.
- Uses MAC address to forward data.

e.g. used in a computer lab and offices to connect PCs

Conclusion:

→ A Switch improves network performance by sending data only to the correct destination devices.

3) Checksum:

→ A checksum is an error-detection technique used to check whether data has been transmitted correctly.

working:

- Sender calculates a value (checksum) from data & sends it with the data,
- Receiver recalculates the ~~checksum~~ checksum.
- If both match, data is correct; otherwise data integrity.

Conclusion:

checksum helps in detecting errors during data transmission & ensures data reliability.

6 Define protocol. why do we need standards? (2079)

→ Protocol:

→ A protocol is a set of rules that defines how data is sent, received and processed between devices in a network. It ensures proper communication between sender and receiver.

eg TCP, IP, HTTP.

Why do we need ~~proper~~ standards?

→ Standards are needed to make sure different devices and network can work together properly.

we need standards because:

- They ensure compatibility between different systems and devices.

- They allow devices from different manufacturers to communicate.
- They make communication reliable & uniform.
- They improve efficiency & reduce error in networking.

e.g. IEEE standards help different network devices work together.

Q Why do we need network topology? Explain star topology along with its merits & demerits.

(2079)

→ Network topology is needed to define how computers and devices are arranged and connected in a network. It helps in planning and managing the network in an efficient way.

We need network topology because

- It helps in easy design and organization of a network.
- It improves performance and communication efficiency.
- It makes troubleshooting easier when problems occur.
- It helps in choosing the best layout based on cost & requirements.

Star topology :

→ In Star topology, all devices are connected to a single central device like Hub or Switch. All communication passes through this central device.

Merits (Advantages)

- Easy to install and manage.
- If one device fails, it does not affect others.
- Easy to detect and fix faults.
- Good performance in small to medium networks.

Demerits (Disadvantages)

- If the central device (Hub/Switch) fails, the whole network stops.
- Requires more cables compared to bus topology.
- Cost is higher due to central device.

8 What is protocol? Explain each layers of OSI model in details. (2079 - 10ng)

⇒ A protocol is a set of rules that defines how data is transmitted, received and processed in a network so that device can communicate properly.

e.g TCP, IP, HTTP.

OSI model:

→ The Open Systems Interconnection (OSI) model was developed by the International Organization for Standardization (ISO) and formalized in 1984. It provides the first framework governing how information should be sent across a network.

→ OSI stands for open system interconnection model. It tells us how data travels in a network. OSI model has 7 layers.

7 Application

6 Presentation

5 Session

4 Transport

3 Network

2 Data link

1 Physical

1) Physical layer:

→ The first layer of OSI model is called Physical layer. The Physical layer describes electrical or optical signals. All the physical connectivity of network takes place such as a cable RJ45. Clamping. It convert data into bits and forward it to the physical layer to the receiving sides.

Devices uses . HUB , ~~SW~~ Repeater .

2) Data link layer:

→ The second layer of the OSI model is called data link layer. The data link layer establishes a connection between two physical connected devices on a network.

→ It is responsible for node-to-node delivery of data. The data link layer consists of two sub layers.

- Logical link control (LLC) sub layer.

- Media access control (MAC) sub layer.

Logical link layer: (LLC):

→ It is responsible for transferring the packets to the network layer of the receiver that is receiving.

medium Access control (MAC):

→ establishes and maintains links between communicating devices

Protocol used this layer are: PPP (point to point protocol), PPTP (point to point tunneling protocol) ARP (address resolution protocol) and RARP (Reverse address resolution protocol)

Devices: Switch

3 network layer:

→ The 3rd layer of OSI network model is called network layer. The network layer is responsible for the source to destination delivery of a packet

1 logical addressing:

→ logical address provides a unique address that identifies computer on internet.

2 Routing:

→ A Router determines the best path to send and receive data from source to destination. protocol used under this layer are

IP, RIP, IGRP, EIGRP, OSPF etc

Devices: Router

4) Transport layer:

→ The 4th layer of OSI model is called transport layer. It is responsible for process-to-process delivery of the entire message.

✓ 1 TCP/IP : it establishes the connection between hosts.

2 UDP : It is unreliable protocol.

The transport layer has the following functionalities:

1 Segmentation

2 Flow Control

3 Error control

4 Sequencing

5) Session layer:

→ The 5th layer of OSI model is called session layer. The session layer is responsible for dialog control, establishing the connection, manage the connection and terminate the connection.

Session layers also control when to send the data in simplex, half duplex and full duplex.

- Full-Duplex
 - Simultaneous two-way communication
- Half-Duplex
 - two way communication 'but not simultaneous'
- Simplex
 - one way communication

6. Presentation layer:

- The 6th layer of OSI model is called presentation layer. This layer is responsible for translation, encryption and compression. The protocol work under this layer is SSL.

7. Application layer:

- This is the final layer of OSI model. This layer provides user interface and supports services such as mail, services, file transfer, browsing internet, remote desktop, connection etc. the protocol that communicate with the application layer are SMTP, HTTP, POP3, IMAP, DHCP, & FTP.